

| Quantity                                     | Item                               | Description                                                                           | Manufacturer | Link                                                                                                                                                | Cost (No Tax/Shipping) |
|----------------------------------------------|------------------------------------|---------------------------------------------------------------------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 12                                           | Servo                              | Powers the wheels and the camera module.                                              | WOWROBO      | <a href="https://shop.wowrobo.com/products/raspi-servo">https://shop.wowrobo.com/products/raspi-servo</a>                                           | \$ 171.88              |
| 1                                            | RPi5/8GB                           | Powers the camera module, controls the rover. The "brains."                           | AAAWave      | <a href="https://aaawave.com/products/raspberry-pi-5-8gb">https://aaawave.com/products/raspberry-pi-5-8gb</a>                                       | \$ 150.00              |
| 1                                            | MicroSD                            | High-powered SD card for video streaming.                                             | Sandisk      | <a href="https://www.sandisk.com/product/sdcard">https://www.sandisk.com/product/sdcard</a>                                                         | \$ 25.99               |
| 1                                            | RPi Active Cooler                  | Provides cooling (improving performance) to the RPi.                                  | Waveshare    | <a href="https://www.waveshare.com/pi5-active-cooler">https://www.waveshare.com/pi5-active-cooler</a>                                               | \$ 4.99                |
| 1                                            | Servo Driver HAT                   | Allows the RPi to control the servos.                                                 |              | <a href="https://www.waveshare.com/bus-servo-driver-hat">https://www.waveshare.com/bus-servo-driver-hat</a>                                         | \$ 18.99               |
| 1                                            | RPi AI Camera                      | Allows the RPi to run object detection/livestreaming.                                 | Adafruit     | <a href="https://www.adafruit.com/product/5147">https://www.adafruit.com/product/5147</a>                                                           | \$ 77.00               |
| 1                                            | Camera Cable - 500mm               | Extends the camera module's range.                                                    |              | <a href="https://www.adafruit.com/product/5148">https://www.adafruit.com/product/5148</a>                                                           | \$ 2.50                |
| 3                                            | US-100 Sensor                      | Allows the RPi to run object avoidance.                                               |              | <a href="https://www.adafruit.com/product/5149">https://www.adafruit.com/product/5149</a>                                                           | \$ 20.85               |
| 1                                            | Stacking Header, GPIO              | Allows the HAT to be attached onto the Pi alongside the Active Cooler.                |              | <a href="https://www.adafruit.com/product/5150">https://www.adafruit.com/product/5150</a>                                                           | \$ 2.95                |
| 12                                           | Male to JST-PH Wires               | Allows for the solderless wiring (acting as a breakout board for power-injection).    | Electromaker | <a href="https://www.electromaker.io/shop/12x1m-male-to-jst-ph-wires">https://www.electromaker.io/shop/12x1m-male-to-jst-ph-wires</a>               | \$ 15.00               |
| 12                                           | Female to JST-PH                   | Same role as the male wires, composite.                                               |              | <a href="https://www.electromaker.io/shop/12x1m-female-to-jst-ph-wires">https://www.electromaker.io/shop/12x1m-female-to-jst-ph-wires</a>           | \$ 15.00               |
| 1                                            | Ball Bearings                      | Facilitate the rocker-bogie's rotational freedom.                                     | Amazon       | <a href="https://www.amazon.com/uxcell-10x5x3mm-10-pack-steel-ball-bearings">https://www.amazon.com/uxcell-10x5x3mm-10-pack-steel-ball-bearings</a> | \$ 7.82                |
| 1                                            | VK-162                             | Allow the RPi to gain GPS-based autonomous capabilities.                              |              | <a href="https://www.amazon.com/Navicat-162-GPS-module">https://www.amazon.com/Navicat-162-GPS-module</a>                                           | \$ 17.98               |
| 1                                            | LiFePO4 12V 15Ah Battery           | Powers the rover.                                                                     |              | <a href="https://www.amazon.com/12-8V-20Ah-LiFePO4-Battery">https://www.amazon.com/12-8V-20Ah-LiFePO4-Battery</a>                                   | \$ 53.19               |
| 1                                            | LiFePO4 Charger                    | Charges the battery.                                                                  |              | <a href="https://www.amazon.com/dp/B078333333">https://www.amazon.com/dp/B078333333</a>                                                             | \$ 25.07               |
| 1                                            | WAGO 221-415 (25 Pack)             | Facilitate the solderless connections with the power gauge wire and the Male/Females. |              | <a href="https://www.amazon.com/WAGO-221-415-25-Pack">https://www.amazon.com/WAGO-221-415-25-Pack</a>                                               | \$ 17.99               |
| 1                                            | Breadboard Jumper Wires Set        | To extend the Male/Female JST-PH wires.                                               |              | <a href="https://www.amazon.com/EDGE-100-Pin-Breadboard-Jumper-Wires-Set">https://www.amazon.com/EDGE-100-Pin-Breadboard-Jumper-Wires-Set</a>       | \$ 8.99                |
| 1                                            | Heat Shrink Tubing Set             | Allow for the solderless collections to be solid.                                     |              | <a href="https://www.amazon.com/Eventronic-Heat-Shrink-Tubing-Set">https://www.amazon.com/Eventronic-Heat-Shrink-Tubing-Set</a>                     | \$ 13.59               |
| 1                                            | Mini Heat Gun                      | Combine with the Heat Shrink Tubing                                                   |              | <a href="https://www.amazon.com/QWORK-Heat-Gun">https://www.amazon.com/QWORK-Heat-Gun</a>                                                           | \$ 8.72                |
| 2                                            | 20A 300W Adjustable Buck Converter | Allows for the power source to safely distribute power to the Pi/Servos.              |              | <a href="https://www.amazon.ca/XLX-High-Power-Adjustable-Buck-Converter">https://www.amazon.ca/XLX-High-Power-Adjustable-Buck-Converter</a>         | \$ 31.98               |
| 1                                            | 20A ANL Fuse                       | Protects the inner circuitry.                                                         |              | <a href="https://www.amazon.com/RENOG-20A-ANL-Fuse">https://www.amazon.com/RENOG-20A-ANL-Fuse</a>                                                   | \$ 12.99               |
| 1                                            | Spade Terminal Connectors          | Allow for a solderless connection to the battery.                                     |              | <a href="https://www.amazon.com/Feggizit-10-Pin-10-Pack">https://www.amazon.com/Feggizit-10-Pin-10-Pack</a>                                         | \$ 7.69                |
| 1                                            | 12 Gauge Wire                      | Allow for power streaming over long-distance power injection.                         |              | <a href="https://www.amazon.com/Voltage-12-Gauge-Wire">https://www.amazon.com/Voltage-12-Gauge-Wire</a>                                             | \$ 11.99               |
| 1                                            | Rod-End Set                        | Allows for the Differential Mechanism to function.                                    |              | <a href="https://www.amazon.com/Que-T-10-Pack">https://www.amazon.com/Que-T-10-Pack</a>                                                             | \$ 9.99                |
| 1                                            | M3 Screw/Nut Set                   | Required for the multipart assembly.                                                  |              | <a href="https://www.amazon.com/Fgruh-10-Pack">https://www.amazon.com/Fgruh-10-Pack</a>                                                             | \$ 21.99               |
| 1                                            | Plumber's Tape                     | Alternative to bushing, bridge the ASA with low frictional losses.                    |              | <a href="https://www.amazon.com/Inches-10-Pack">https://www.amazon.com/Inches-10-Pack</a>                                                           | \$ 7.39                |
| 1                                            | Heat-Set Inserts                   | Required for the multipart assembly.                                                  |              | <a href="https://www.amazon.com/Thread-10-Pack">https://www.amazon.com/Thread-10-Pack</a>                                                           | \$ 6.27                |
| 1                                            | Soldering Tin                      | Required for the stacking header soldering.                                           | Amazon       | <a href="https://www.amazon.com/MAIYU-10-Pack">https://www.amazon.com/MAIYU-10-Pack</a>                                                             | \$ 8.99                |
| 1                                            | Soldering Iron                     | Required for the Heat-Set Inserts and the Soldering.                                  | Aliexpress   | <a href="https://www.aliexpress.us/item/3256814892227202.html">https://www.aliexpress.us/item/3256814892227202.html</a>                             | \$ 31.38               |
| Total:                                       |                                    |                                                                                       |              |                                                                                                                                                     | \$ 809.16              |
| Total (*with taxes and shipping):            |                                    |                                                                                       |              |                                                                                                                                                     | \$ 1,000.00            |
| Total (*with taxes and shipping AND prints): |                                    |                                                                                       |              |                                                                                                                                                     | \$ 1,350.00            |